

Joe Youssouf

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Summary

Data scientist with 6+ years of experience developing statistical evaluation frameworks, metrics, and production ML systems for high-frequency sensor and vehicle telemetry data. Experienced measuring model and system performance, investigating anomalies, estimating rare operational events, and supporting deployment-readiness decisions in reliability-critical environments. Strong background partnering with engineering and product teams to translate ambiguous operational data into safety, quality, and reliability decisions.

Industry Experience

Oden Technologies

New York, NY

Senior Data Scientist

Oct. 2024 - Present

Build statistical evaluation frameworks and production ML systems for high-frequency manufacturing sensor data, supporting anomaly detection, offline quality prediction, and operator decision-making across customer manufacturing environments.

- **ProcessAI:** Designed and deployed a generalized anomaly detection system across 16 customer manufacturing sites with hundreds of high-frequency sensor signals per site; developed evaluation metrics to identify production-impacting process anomalies and supported a copper rod mill deployment that reported a 22% OEE improvement over the following 8 months.
- **Predictive Quality:** Built models that predicted offline quality outcomes from live sensor data and informed ProcessAI's multi-model recommendation logic, helping operators adjust production toward better quality and throughput before final measurements were available.
- **Oden Copilot:** Built MCP tools and Gemini-based agents for an internal conversational analytics chatbot that enabled users to query real-time and historical manufacturing data, navigate the platform, and generate dashboards and custom reports.

Preteckt, acquired by Diesel Laptops, now Watchtower

Hamilton, ON

Data Scientist, Product

Sept. 2020 - Oct. 2024

Built production ML and statistical evaluation systems for predictive maintenance using high-frequency vehicle telemetry, diagnostic sensor data, and fault codes from thousands of heavy-duty commercial vehicles across North America. Delivered human-in-the-loop maintenance alerts and repair recommendations in production operating environments.

- Trained, deployed, and monitored 80+ predictive maintenance models across vehicle classes, makes, and model years using telemetry, fault codes, and diagnostic sensor data.
- Developed model monitoring and evaluation pipelines to investigate performance anomalies, quantify model drift, and support deployment-readiness decisions across production models.

- Delivered measured customer outcomes including a 43% increase in maintenance productivity and 24% reduction in material costs at New York City MTA, plus a 50% reduction in labor hours at Saskatoon Transit.
- Helped scale a predictive maintenance analytics platform from 100 to 15,000+ connected vehicles across North American trucking, transit, and passenger vehicle fleets by building scalable model evaluation, monitoring, and deployment workflows.

Gordie Howe International Bridge

Windsor, ON

Data Scientist Intern, Analytics

Jan. 2020 - Sept. 2020

- Conducted population density and urban growth analysis that identified an access-route planning issue and helped avoid over \$85,000 in construction costs.
- Designed stakeholder-facing data visualizations for internal planning and external communication.

ENWIN Utilities

Windsor, ON

Data Scientist Intern, Analytics

Jan. 2019 - Sept. 2019

- Designed ARIMA forecasting models for smart-grid optimization, reducing excess off-peak power waste by up to 8%.
- Developed a monthly power metrics dashboard in Grafana that was regularly presented to provincial stakeholders and used for operational decision-making.

Education

University of Windsor

Windsor, ON

BASc, Honours Electrical Engineering

- Dean's Honour Roll, Minor in Mathematics and Statistics

Technical Skills

Statistical & ML Methods: Statistical evaluation, metric design, anomaly detection, rare-event evaluation, time-series analysis, predictive modeling, model monitoring, drift detection

Programming Languages: Python, SQL, Go, Bash, HTML/CSS

Data Science Frameworks/Packages: Pandas, Scikit-learn, NumPy, Matplotlib, Django/DRF, Flask

MLOps & Data Engineering: MLflow, Airflow, Dagster, dbt, Apache Beam/Dataflow, PostgreSQL, ClickHouse, TimescaleDB

Tools & Environments: Git, Google Cloud Platform, Tableau, Jupyter, MacOS, Linux

LLMs & Agents: Model Context Protocol (MCP), Gemini agents, tool calling, structured data retrieval, natural-language analytics, dashboard/report generation